

ECE 3436

EXAM 1 PROGRAMMING

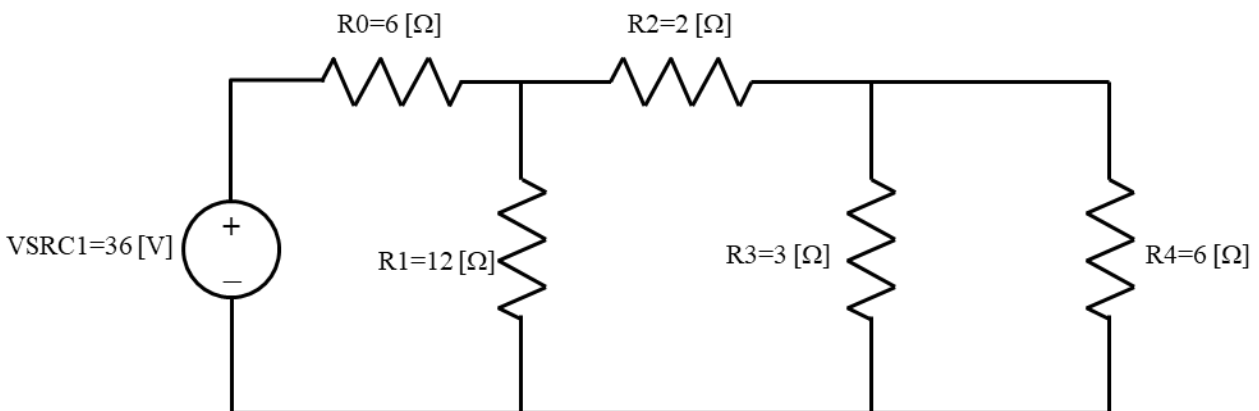
Programming

Download the starter project named Exam_1_project.zip from Canvas or Teams.

Refer to the following circuit schematic to complete the programming tasks. After you have completed tasks 0-3, have someone from the staff check your work. Then move on to the next tasks. Ask for guidance from the staff if you get stuck.

IMPORTANT:

- You are not allowed to use immediate values in your code after Tasks 2 (#VSR1 is allowed). You must use the registers. Also, once you assign a value to register in a Task you are not allowed to alter it after the Task.
- Place the code for each task in the designated area in the main.s file
- Carefully arrange the Registers used in each Task. Sometimes, changing the orders in the task may be helpful. For example, in Task 3, you may need additional registers to calculate R5 but you don't need additional registers to calculate ISRC1 in R8, you may use Register R8 to help calculate R5 first, then calculate the ISRC1 and save it to register R8. You can also use the register have not been assigned, like using Register R12 in Task 3.



TASK 0

Write an EQU statement equating the label VSR1 with the decimal number 36. Also, export the label "Start". The code will not build/assemble until you complete this task.

TASK 1

Write the directive line to create a code area named `|.text|`. The area should be read-only and start at a memory address that is a multiply of 4.

TASK 2

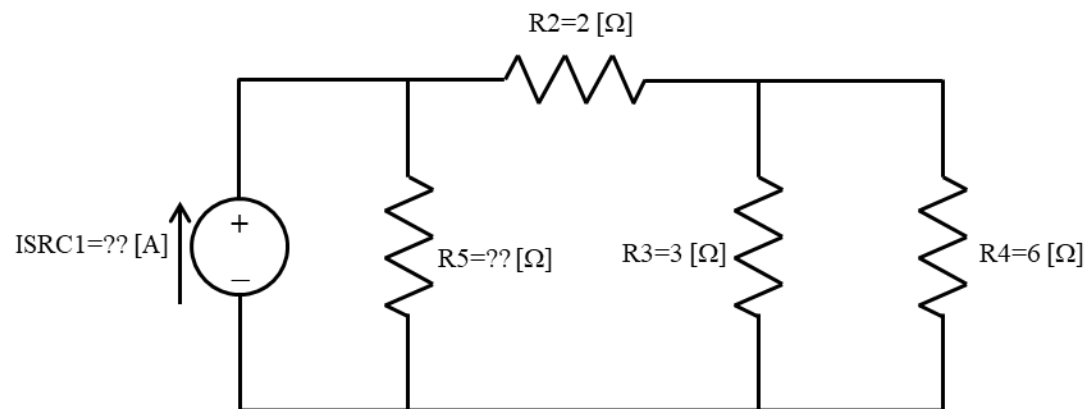
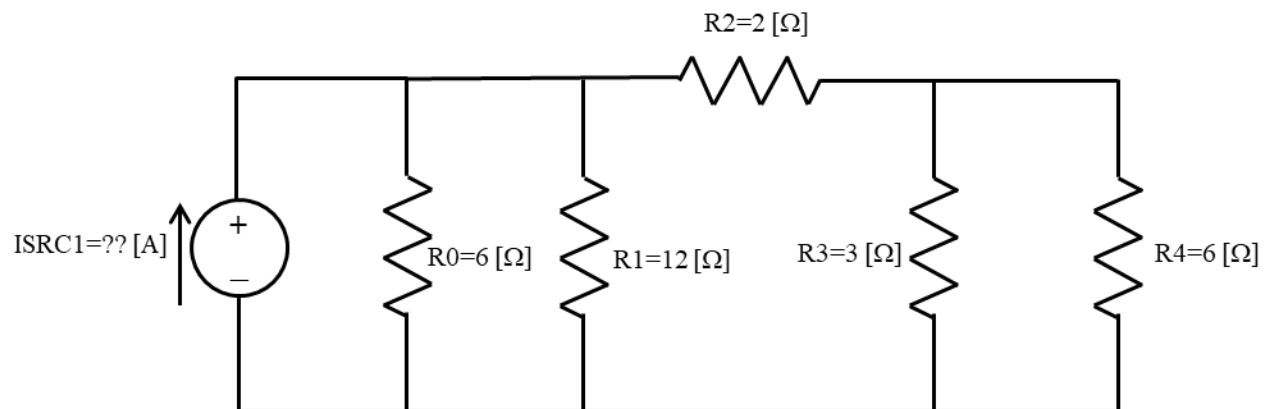
Load register **R0** with the resistor value R0. Load register **R1** with the resistor value R1. Load register **R2** with the resistor value R2. Load register **R3** with the resistor value R3. Load register **R4** with the resistor value R4. (MOV or LDR are both OK)

TASK 3

The first diagram below shows the circuit after a source transformation on the original circuit. The second, after finding an equivalent resistance.

Calculate the value of $ISRC1 (=VSRC1/R0)$ and place the value in register **R8**.

Calculate the value of R5 in ohms and place the value in register **R5**.

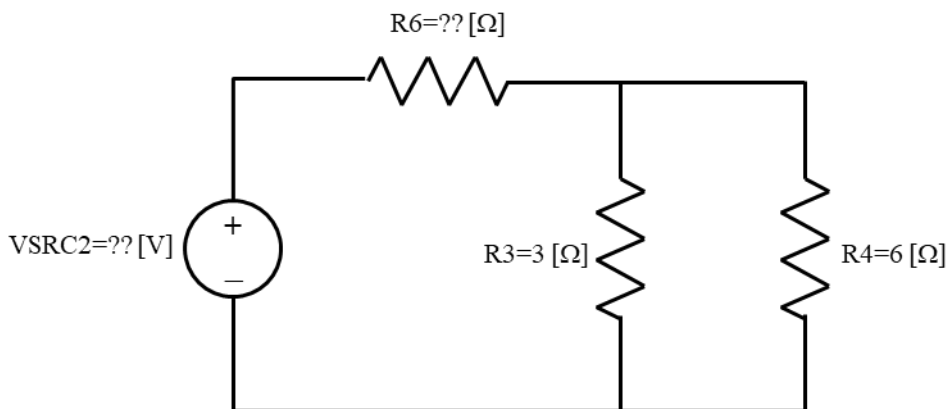
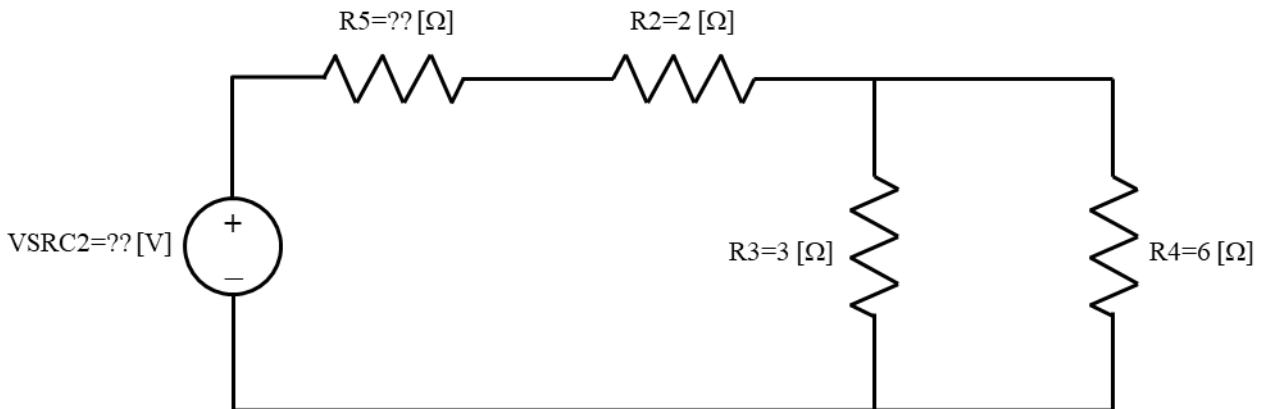


TASK 4

The following two diagrams show the circuit after a source transformation and finding an equivalent resistance.

Calculate the value of V_{SRC2} ($=I_{SRC1} \cdot R_5$) and place the value in register **R9**.

Calculate the value of R_6 in ohms and place the value in register **R6**.

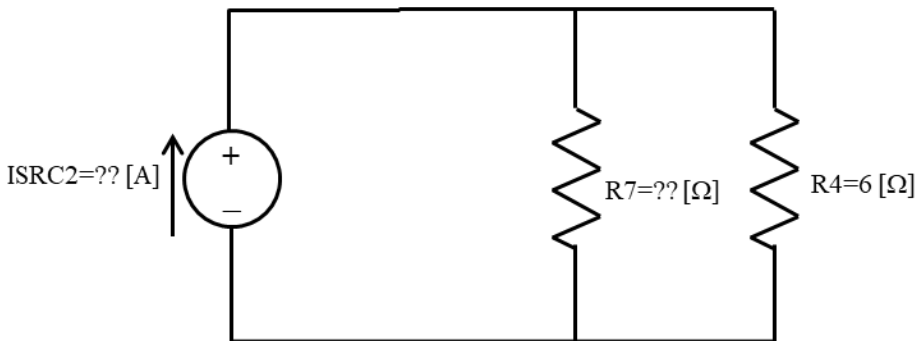
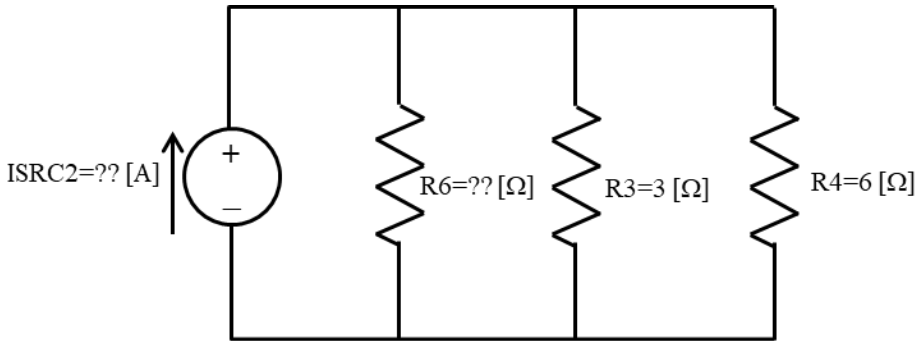


TASK 5

The following two diagrams show the circuit after a source transformation and finding an equivalent resistance.

Calculate the value of I_{SRC2} ($=V_{SRC2}/R_6$) and place the value in register **R10**.

Calculate the value of R_7 in ohms and place the value in register **R7**.

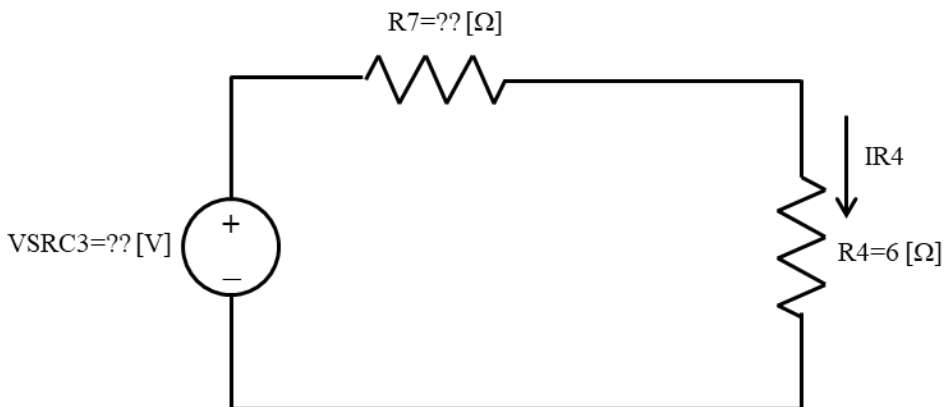


TASK 6

The following two diagrams show the circuit after a source transformation and finding an equivalent resistance.

Calculate the value of $VSRC3 (=ISRC2 \cdot R7)$ and place the value in register **R11**.

Calculate the current $IR4$ and place the result in register **R12**.



TASK 7

Change the value of VSRC1 in the EQU statement to 72 and verify that your answers are still correct.

Check point

At the end of your program, you should have the following information in your registers:

Register	Value
R0	R0
R1	R1
R2	R2
R3	R3
R4	R4
R5	R5
R6	R6
R7	R7
R8	ISRC1
R9	VSRC2
R10	ISRC2
R11	VSRC3
R12	IR4
R13	RESERVED
R14	RESERVED
R15	RESERVED